

David Burstein

The Shmunis School of Biomedicine and Cancer Research
George S. Wise Faculty of Life Sciences, Tel Aviv University

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Lab website: <https://burstein-lab.github.io/>

I study interactions within microbial communities using integrated computational and experimental approaches. My work centers on CRISPR-Cas systems, mobile genetic elements, and antimicrobial resistance, exploring both fundamental questions and translational applications in medicine and biotechnology.

Personal Details

Citizenship: Israeli/French

Employment and Positions

- 2024 – Associate Professor, The Shmunis School of Biomedicine and Cancer Research
George S. Wise Faculty of Life Sciences, Tel Aviv University
- 2025 – Visiting Professor (Sabbatical), Department of Microbiology & Immunology, UBC
- 2018 – 2023 Senior Lecturer (Assistant Professor equiv.), The Department of Molecular Microbiology and Biotechnology, George S. Wise Faculty of Life Sciences, Tel Aviv University
- 2015 – 2018 Postdoctoral fellow, The California Institute for Quantitative Biosciences, University of California, Berkeley
(Advisor: Jennifer A. Doudna)
- 2013 – 2015 EMBO fellow, Earth and Planetary Sciences, University of California, Berkeley
(Advisor: Jill F. Banfield)

Education

- 2008 – 2013 PhD., Department of Cell Research and Immunology, Tel Aviv University
(Advisor: Tal Pupko)
- 2006 – 2008 MSc., Department of Cell Research and Immunology, Tel Aviv University
(Advisor: Tal Pupko)
- 2001 – 2004 BSc. Double major in Life Sciences and Computer Science, Tel Aviv University

Publications

Metrics (as of September 2026):

h-index: 38 ([Google Scholar](#)), 32 ([Scopus](#))

Total citations: 9,265 ([Google Scholar](#)), 6,237 ([Scopus](#))

Self in bold; Students/personnel under my supervision underlined

* Equal contribution

✉ Corresponding author

Peer-Reviewed Publications

1. Rannon E., **Burstein, D**[✉]. Optimizing Protein Tokenization: Reduced Amino Acid Alphabets for Efficient and Accurate Protein Language Models. **Bioinformatics**. 2026, July; Vol. 42: btag226
2. Fraikin, N., Samuel, B., **Burstein, D**[✉], Lesterlin, C[✉]. Strategies for zygotic gene expression during plasmid establishment. **Plasmid**. 2025, June; Vol. 134: pp. 102754
3. Rannon E., Shaashua, S., **Burstein, D**[✉]. DRAMMA: A multifaceted machine learning approach for novel antimicrobial resistance gene detection in metagenomic data. **Microbiome**. 2025, March; Vol. 13: pp. 67
4. Samuel, B., Mittelman, K., Croitoru, S. Y., Ben-Haim, M., **Burstein, D**[✉]. Diverse anti-defense systems are encoded in the leading region of plasmids. **Nature**. 2024, October; Vol. 635: pp. 186–192.
5. Hershko Y., Rannon E., Adler A., **Burstein D.**, and Barkan D. WarA, a remote *Nocardia wallacei* homolog of NpmA and KamB, confers broad spectrum aminoglycoside resistance in *Nocardia* and *Mycobacteria*. **International Journal of Antimicrobial Agents**. 2024, February; Vol: 63(2), 107089.
6. Miller, D., Arias, O., **Burstein, D**[✉]. GeNLP: An interactive web application for microbial gene exploration and prediction. **Bioinformatics**. 2024, January; Vol: 40(2), btae034.
7. Harari, S., Miller, D., Fleishon, S., **Burstein, D.**, and Stern, A. Using big sequencing data to identify chronic SARS-Coronavirus-2 infections. **Nature Communications**. 2024, January; Vol. 15: pp. 648.
8. Alon, D. M.*^{*}, Mittelman, K.*, Stibbe, E., Countryman, S., Stodieck, L., Doraisingam, S., Leal Martin, D. M., Hamo, E. R., Pines, G[✉], **Burstein, D**[✉]. CRISPR-based genetic diagnostics in microgravity. **Biosensors & Bioelectronics**. 2023, October; Vol: 237, 115479
9. Hershko, Y., Levytskyi, K., Rannon, E., Assous, M. V., Ken-Dror, S., Amit, S., Ben-Zvi, H., Sagi, O., Schwartz, O., Sorek, N., Szwarcwort, M., Barkan, D., **Burstein, D.**, and Adler, A. Phenotypic

and genotypic analysis of antimicrobial resistance in *Nocardia* species. ***Journal of Antimicrobial Chemotherapy***. 2023, September; Vol: 78(9): pp. 2783.

10. Miller, D., Stern, A., **Burstein, D.**[✉]. Deciphering microbial gene function using natural language processing. ***Nature Communications***. 2022, September; Vol. 13: pp. 5731.
11. Alon, D. M., Partosh, T., **Burstein, D.**, Pines, G. Rapid and sensitive on-site genetic diagnostics of pest fruit flies using CRISPR-Cas12a. ***Pest Management Science***. 2023, January; Vol. 79(1): pp. 68–75.
12. Tsuchida, C. A., Zhang, S., Doost, M. S., Zhao, Y., Wang, J., O'Brien, E., Fang, H., Li, C. P., Li, D., Hai, Z. Y., Chuck, J., Brötzmann, J., Vartoumian, A., **Burstein, D.**, Chen, X. W., Nogales, E., Doudna, J. A., Liu J. J. G. Chimeric CRISPR-CasX enzymes and guide RNAs for improved genome editing activity. ***Molecular Cell***. 2022, March; Vol. 82: pp. 1199–1209.
13. Nahmad, A. D., Raviv, Y., Horovitz-Fried, M., Sofer, I., Akriv, T., Nataf, D., Dotan, I., Carmi, Y., **Burstein, D.**, Wine, Y., Benhar, I., Barzel, A. Engineered B cells expressing an anti-HIV antibody enable memory retention, isotype switching and clonal expansion. ***Nature Communications***. 2020, November; Vol. 11: pp. 5851.
14. Harrington, L. B., Ma, E., Chen, J. S., Witte, I. P., Gertz, D., Paez-Espino, D., Al-Shayeb, B., Kyrpides, N. C., **Burstein, D.**, Banfield, J. F., and Doudna, J. A. A scoutRNA is required for some Type V CRISPR-Cas systems. ***Molecular Cell***. 2020, July; Vol. 79: pp. 416–424.e5
15. Méheust, R., **Burstein, D.**, Castelle, C. J., and Banfield, J. F. The distinction of CPR bacteria from other bacteria based on protein family content. ***Nature Communications***. 2019, September; Vol. 10: pp. 4173.
16. Diamond, S., Andeer, P. F., Li, Z., Crits-Christoph, A., **Burstein, D.**, Anantharaman, K., Lane, K. R., Thomas, B. C., Pan, C., Northen, T. R., and Banfield, J. F. Mediterranean grassland soil C–N compound turnover is dependent on rainfall and depth, and is mediated by genomically divergent microorganisms. ***Nature Microbiology***. 2019, August; Vol. 4: pp. 1356–1367.
17. Wright, A. V., Wang, J. Y., **Burstein, D.**, Harrington, L. B., Paez-Espino, D., Kyrpides, N. C., Iavarone, A. T., Banfield, J. F., and Doudna, J. A. A functional mini-Integrase in a two-protein type V-C CRISPR system. ***Molecular Cell***. 2019, February; Vol. 73: pp. 727–737.e3.
18. Carnevali, P. B. M., Schulz, F., Castelle, C. J., Kantor, R. S., Shih, P. M., Sharon, I., Santini, J. M., Olm, M. R., Amano, Y., Thomas, B. C., Anantharaman, K., **Burstein, D.**, Becraft, E. D., Stepanauskas, R., Woyke, T., and Banfield, J. F. Hydrogen-based metabolism as an ancestral trait in lineages sibling to the Cyanobacteria. ***Nature Communications***. 2019, January; Vol. 10: pp. 463.
19. Harrington, L. B.* , **Burstein, D.***, Chen, J. S., Paez-Espino, D., Ma, E., Witte, I. P., Cofsky, J. C., Kyrpides, N. C., Banfield, J. F., and Doudna, J. A. Programmed DNA destruction by miniature

CRISPR-Cas14 enzymes. **Science**. 2018, November; Vol. 362: pp. 839–842.

20. Dudek, N. K., Sun, C. L., **Burstein, D.**, Kantor, R. S., Aliaga Goltsman, D. S., Bik, E. M., Thomas, B. C., Banfield, J. F., and Relman, D. A. Novel microbial diversity and functional potential in the marine mammal oral microbiome. **Current Biology**. 2017, December; Vol. 27: pp. 3752-3762.e6.
21. Hernsdorf, A. W., Amano, Y., Miyakawa, K., Ise, K., Suzuki, Y., Anantharaman, K., Probst, A., **Burstein, D.**, Thomas, B. C., and Banfield, J. F. Potential for microbial H₂ and metal transformations associated with novel bacteria and archaea in deep terrestrial subsurface sediments. **The ISME Journal**. 2017, August; Vol. 11: pp. 1915–1929.
22. Paul, B. G., **Burstein, D.**, Castelle, C. J., Handa, S., Arambula, D., Czornyj, E., Thomas, B. C., Ghosh, P., Miller, J. F., Banfield, J. F., and Valentine, D. L. Retroelement-guided protein diversification abounds in vast lineages of Bacteria and Archaea. **Nature Microbiology**. 2017, June; Vol. 2: pp. 17045.
23. East-Seletsky, A., O'Connell, M. R., **Burstein, D.**, Knott, G. J., and Doudna, J. A. RNA Targeting by functionally orthogonal type VI-A CRISPR-Cas enzymes. **Molecular Cell**. 2017, May; Vol. 66: pp. 373-383.e3.
24. Olm, M. R., Brown, C. T., Brooks, B., Firek, B., Baker, R., **Burstein, D.**, Soenjoyo, K., Thomas, B. C., Morowitz, M., and Banfield, J. F. Identical bacterial populations colonize premature infant gut, skin, and oral microbiomes and exhibit different in situ growth rates. **Genome Research**. 2017, April; Vol. 27: pp. 601–612.
25. **Burstein, D.***, Harrington, L. B.*, Strutt, S. C., Probst, A. J., Anantharaman, K., Thomas, B. C., Doudna, J. A., and Banfield, J. F. New CRISPR–Cas systems from uncultivated microbes. **Nature**. 2017, February; Vol. 542: pp. 237–241.
26. Probst, A. J., Castelle, C. J., Singh, A., Brown, C. T., Anantharaman, K., Sharon, I., Hug, L. A., **Burstein, D.**, Emerson, J. B., Thomas, B. C., and Banfield, J. F. Genomic resolution of a cold subsurface aquifer community provides metabolic insights for novel microbes adapted to high CO₂ concentrations. **Environmental Microbiology**. 2017, February; Vol. 19: pp. 459–474.
27. East-Seletsky, A.*, O'Connell, M. R.*, Knight, S. C., **Burstein, D.**, Cate, J. H. D., Tjian, R., and Doudna, J. A. Two distinct RNase activities of CRISPR-C2c2 enable guide-RNA processing and RNA detection. **Nature**. 2016, October; Vol. 538: pp. 270–273.
28. **Burstein, D.**, Sun, C. L., Brown, C. T., Sharon, I., Anantharaman, K., Probst, A. J., Thomas, B. C., and Banfield, J. F. Major bacterial lineages are essentially devoid of CRISPR-Cas viral defence systems. **Nature Communications**. 2016, February; Vol. 7: pp. 10613.
29. **Burstein, D.**, Amaro, F., Zusman, T., Lifshitz, Z., Cohen, O., Gilbert, J. A., Pupko, T., Shuman, H. A., and Segal, G. Genomic analysis of 38 *Legionella* species identifies large and diverse effector

repertoires. **Nature Genetics**. 2016, February; Vol. 48: pp. 167–175.

30. Teper, D.* , **Burstein, D.***, Salomon, D., Gershovitz, M., Pupko, T., and Sessa, G. Identification of novel *Xanthomonas euvesicatoria* type III effector proteins by a machine-learning approach. **Molecular Plant Pathology**. 2016, April; Vol. 17: pp. 398–411.
31. Anantharaman, K., Brown, C. T., **Burstein, D.**, Castelle, C. J., Probst, A. J., Thomas, B. C., Williams, K. H., and Banfield, J. F. Analysis of five complete genome sequences for members of the class Peribacteria in the recently recognized Peregrinibacteria bacterial phylum. **PeerJ**. 2016, January; Vol. 4: pp. e1607.
32. **Burstein, D.***, Satanower, S.* , Simovitch, M.* , Belnik, Y., Zehavi, M., Yerushalmi, G., Ben-Aroya, S., Pupko, T., and Banin, E. Novel type III effectors in *Pseudomonas aeruginosa*. **mBio**. 2015, May; Vol. 6: pp. e00161-15.
33. Sharon, I., Kertesz, M., Hug, L. A., Pushkarev, D., Blauwkamp, T. A., Castelle, C. J., Amirebrahimi, M., Thomas, B. C., **Burstein, D.**, Tringe, S. G., Williams, K. H., and Banfield, J. F. Accurate, multi-kb reads resolve complex populations and detect rare microorganisms. **Genome Research**. 2015, April; Vol. 25: pp. 534–543.
34. Lifshitz, Z., **Burstein, D.**, Schwartz, K., Shuman, H. A., Pupko, T., and Segal, G. Identification of novel *Coxiella burnetii* Icm/Dot effectors and genetic analysis of their involvement in modulating a mitogen-activated protein kinase pathway. **Infection and Immunity**. 2014, September; Vol. 82: pp. 3740–3752.
35. Yosef, I., Shitrit, D., Goren, M. G., **Burstein, D.**, Pupko, T., and Qimron, U. DNA motifs determining the efficiency of adaptation into the *Escherichia coli* CRISPR array. **Proceedings of the National Academy of Sciences**. 2013, August; Vol. 110: pp. 14396–14401.
36. Cohen, O., Ashkenazy, H., Levy Karin, E., **Burstein, D.**, and Pupko, T. CoPAP: Coevolution of presence-absence patterns. **Nucleic Acids Research**. 2013, July; Vol. 41: pp. W232–W237.
37. Lifshitz, Z.* , **Burstein, D.***, Peeri, M., Zusman, T., Schwartz, K., Shuman, H. A., Pupko, T., and Segal, G. Computational modeling and experimental validation of the *Legionella* and *Coxiella* virulence-related type-IVB secretion signal. **Proceedings of the National Academy of Sciences**. 2013, February; Vol. 110: pp. E707–E715.
38. Cohen, O., Ashkenazy, H., **Burstein, D.**, and Pupko, T. Uncovering the co-evolutionary network among prokaryotic genes. **Bioinformatics**. 2012, September; Vol. 28: pp. i389–i394.
39. Amit, M., Donyo, M., Hollander, D., Goren, A., Kim, E., Gelfman, S., Lev-Maor, G., **Burstein, D.**, Schwartz, S., Postolsky, B., Pupko, T., and Ast, G. Differential GC content between exons and introns establishes distinct strategies of splice-site recognition. **Cell Reports**. 2012, May; Vol. 1: pp. 543–556.

40. **Burstein, D.***, Gould, S. B.*, Zimorski, V., Kloesges, T., Kiosse, F., Major, P., Martin, W. F., Pupko, T., and Dagan, T. A machine learning approach to identify hydrogenosomal proteins in *Trichomonas vaginalis*. **Eukaryotic Cell**. 2012, February; Vol. 11: pp. 217–228.
41. Gelfman, S., **Burstein, D.**, Penn, O., Savchenko, A., Amit, M., Schwartz, S., Pupko, T., and Ast, G. Changes in exon–intron structure during vertebrate evolution affect the splicing pattern of exons. **Genome Research**. 2012, January; Vol. 22: pp. 35–50.
42. Barzel, A., Privman, E., Peeri, M., Naor, A., Shachar, E., **Burstein, D.**, Lazary, R., Gophna, U., Pupko, T., and Kupiec, M. Native homing endonucleases can target conserved genes in humans and in animal models. **Nucleic Acids Research**. 2011, August; Vol. 39: pp. 6646–6659.
43. **Burstein, D.**, Zusman, T., Degtyar, E., Viner, R., Segal, G., and Pupko, T. Genome-scale identification of *Legionella pneumophila* effectors using a machine learning approach. **PLoS Pathogens**. 2009, July; Vol. 5: pp. e1000508.
44. Schwartz, S., Silva, J., **Burstein, D.**, Pupko, T., Eyra, E., and Ast, G. Large-scale comparative analysis of splicing signals and their corresponding splicing factors in eukaryotes. **Genome Research**. 2008, January; Vol. 18: pp. 88–103.
45. Ulitsky, I., **Burstein, D.**, Tuller, T., and Chor, B. The average common substring approach to phylogenomic reconstruction. **Journal of Computational Biology**. 2006, March; Vol. 13: pp. 336–350.

Peer-Reviewed Proceedings

46. **Burstein, D.**, Ulitsky, I., Tuller, T., and Chor, B. Information theoretic approaches to whole genome phylogenies. *Journal of Computational Biology*. **International Conference on Research in Computational Molecular Biology (RECOMB)**. Cambridge, MA, USA. 2005.

Invited Highlight Papers (Non-Peer-Reviewed)

47. Mittelman, K., and **Burstein, D**✉. Tiny hidden genes within our microbiome. **Cell**. 2019, August; Vol. 178: pp. 1034–1035

Papers under review

48. Shur G., **Burstein, D**✉. FAMUS: A Few-Shot Learning Framework for Large-Scale Protein Annotation. 2026, *under revision in Bioinformatics*.
49. Virolle, C., Ferrarin, S., Panis, G., Baffert, Y., Dedieu-Berne, A., Guerin, J., Cayron, J., Traore, D.A.K., Martinez-Absalon, S., Zenati, R., Delolme, F., Page, A., Bigot, S., Yamaichi, Y., Lopatkin, A.J., Viollier, P.H., **Burstein, D.**, Terradot, L., Lesterlin, C. Plasmid-encoded host reprogramming promotes plasmid dissemination. 2026, *under revision in Nature Microbiology*.

50. Rannon, E., **Burstein, D**[✉]. Leveraging Natural Language Processing to Unravel the Mystery of Life: A Review of NLP Approaches in Genomics, Transcriptomics, and Proteomics. 2026, *under review in Genome Biology*.
51. Solan, Y., Shur, G., **Burstein, D**[✉]. Global-scale analysis of the taxonomic and ecological distribution of key pathogenic secretion systems. 2026, *under revision in mSystems*.

Patents

Granted Patents:

- 2025 US12227753B2: CasY compositions and methods of use, *The Regents of the University of California*.
- 2025 US12264314B1: CasZ compositions and methods of use, *The Regents of the University of California*.
- 2024 US11970719B2: Class 2 CRISPR/Cas compositions and methods of use, *The Regents of the University of California*.
- 2023 US11795472B2: RNA-guided nucleic acid modifying enzymes and methods of use thereof, *The Regents of the University of California*.
- 2023 GB2582100B: Cas12c Compositions and methods of use, *The Regents of the University of California*.

Pending Patents:

- 2024 WO2024166096A1 Methods and Constructs for Improving Conjugation Efficiency, *Ramot at Tel Aviv University Ltd*.

Grant Funding

- 2025 European Research Council (ERC) Consolidator Grant, “*Systematic Investigation of Anti-Defense Function and Regulation in Bacterial Conjugation*” (DEFCONJ).
PI, 60 months, 2,275,000 Euro (~3,680,000 CAD)
- 2025 Ministry of Defence, “An LLM-based approach to predict the function of genes and design tailored pathways and genomic segments”
Co-PI, 12 months, 600,000 NIS (~264,000 CAD)
- 2024 TAU Center for Artificial Intelligence & Data Science (TAD), “*Genomic Language Models Integrating Hierarchical Biological Knowledge*”.
PI, 12 months 120,000 NIS (~52,000 CAD)
- 2023 Israel Science Foundation (ISF) “*Natural language models for predicting gene function in microbes*”.
PI, 60 months, 1,350,000 NIS (~595,000 CAD)
- 2023 Ministry of Defence, “*Language model approaches for detection of genome editing events*”.
PI, 24 months, 572,000 NIS (~252,000 CAD)
- 2020 Ministry of Defence, “*Detection of Antimicrobial Resistance Genes using Computational Approaches and Metagenomic Data Analysis*”.
PI, 36 months, 800,000 NIS (~353,000 CAD)
- 2020 Israel Innovation Authority, “*CRISPR-IL consortium: AI-based Infrastructure for Improving Accuracy and Efficiency of Genome Editing using CRISPR Technology*”.

Co-PI, part of a joint academic and industry consortium, 18 months, 37 million NIS (~16.3 million CAD), out of which 500,000 NIS to my lab (~220,000 CAD)

- 2019 Mammoth Biosciences (Sponsored Research) , *“Expansion of class 2 CRISPR-Cas systems repertoire”*.
PI, 30 months, 588,000 USD (~818,000 CAD)
- 2018 Israel Council for Higher Education, *“Metagenomic and Machine Learning Approaches for Discovery of CRISPR Systems and Resistance Genes”*.
PI, 170,000 NIS (~75,000 CAD)
- 2018 Israel Science Foundation (ISF), *“Discovery and characterization of novel CRISPR-Cas systems”*.
PI, 60 months, 1,675,000 NIS (~739,000 CAD)

Awards and Fellowships

- 2018 Alon Fellowship for Outstanding Young Faculty, Israel Council for Higher Education
- 2014 Long-Term Postdoctoral Fellowship, European Molecular Biology Organization (EMBO)
- 2013 Rothschild Postdoctoral Fellowship, Yad Hanadiv
- 2012 Prof. Willy Hirsch Prize for Original Research in Microbiology, Israel Society for Microbiology
- 2009 Excellence Fellowship, Converging Technologies Program, Israeli Council for Higher Education (48 months)
- 2008 Wolf Foundation Award for excellent MSc students
- 2006 Excellence Fellowship, Edmond J. Safra Bioinformatics Center (36 months)

Presentations (2018 onward)

- 2027 EMBO Symposium, Immune System of Bacteria, University of Cambridge. *Invited Speaker*
- 2026 Gordon Research Conference, Microbiome Editing, Pomona. *Invited Speaker*
Conference Symposium, CSHL meeting on Microbiomes, Cold Spring Harbor. *Invited Speaker*
Conference Symposium, ILANIT/FISEB Experimental Biology. Eilat. *Invited Speaker*
- 2025 Conference Symposium, Israel Data Science and AI Initiative annual meeting. *Invited Speaker*
Conference Symposium, Center for AI and Data Science, Tel Aviv University. *Invited Speaker*
Seminar, Department of Microbiology & Immunology, University of British Columbia
Seminar, The Shmunis School of Biomedicine and Cancer Research, Tel Aviv University
Seminar, Department of Biochemistry & Molecular Biology, Tel Aviv University
- 2024 Conference Symposium, Doudna Lab 30 Year Symposium. UC Berkeley. *Invited Speaker*
Conference Symposium, UC Berkeley-TAU Initiative in Computational Biology and Bioinformatics. Center for Regulatory Genomics (CRG), Barcelona. *Invited Speaker*

Seminar, Department of Life Sciences, Ben-Gurion University

- 2023 Conference Symposium, Bacterial Genetics and Ecology, Copenhagen. *Speaker*
Conference Symposium, Israeli Bioinformatics Symposium. *Invited Speaker*
EMBO Symposium, Immune System of Bacteria. Weizmann Institute. *Speaker*
Conference Symposium, ILANIT Experimental Biology Conference. Eilat. *Speaker*
Seminar, Department of Microbiology, Immunology, and Genetics, Ben-Gurion University
Seminar, The Mina and Everard Goodman Faculty of Life Sciences, Bar-Ilan University
Seminar, Plant & Environmental Sciences. Weizmann Institute of Science
Seminar, Plant and Environmental Sciences, Hebrew University of Jerusalem
- 2021 Conference Symposium, Paradigm Changes in the Management of Early Pregnancy and
Pregnancy Loss. Dubai (delivered virtually). *Invited Speaker*
Online Workshop, Bioinformatic tools for biological insight. University of Bergen.
Invited Speaker
Seminar, Faculty of Agriculture, Food and Environment, The Hebrew University of Jerusalem
Seminar, Israel Institute for Biological Research
Seminar, The Horizon community of educators, Israel Space Agency
Seminar, Google Israel
- 2020 Online Symposium, Israeli Society of Evolutionary Biology annual meeting. *Invited Speaker*
Conference Symposium, Synthetic Biology. Israel Institute for Biological Research
Invited Speaker
Seminar, School of Computer Science and Engineering, The Hebrew University of Jerusalem
- 2019 Conference Symposium, Harvey Prize Symposium, Technion. *Invited Speaker*
Conference Symposium, UC Berkeley-TAU Initiative in Computational Biology and
Bioinformatics. Tel Aviv University. *Invited Speaker*
Online Workshop, Microbiome Metagenomics. College of Health Sciences, Uganda Virus
Research Institute, Makerere University. *Invited Speaker*
Conference Symposium, The Israeli Society for Microbiology Meeting. Weizmann
Institute of Science. *Invited Speaker*
Conference Symposium, Annual Broad-ISF Symposium, Jerusalem. *Invited Speaker*
Conference Symposium, Workshop on Nanoscience and Nanotechnology. Center for
Nanoscience & Nanotechnology, Tel Aviv University. *Invited Speaker*
Seminar, Gynaecology and Maternity Center, Sheba Hospital
Seminar, Institute of Evolution, University of Haifa
- 2018 Conference Symposium, UC Berkeley-TAU Initiative in Computational Biology and
Bioinformatics. UC Berkeley. *Invited Speaker*

Conference Symposium, The Israeli Society for Microbiology Meeting. Ben Gurion University
Invited Speaker
Seminar, Faculty of Life Science, Bar Ilan University
Seminar, Department of Life Science, Ben Gurion University

Supervision

I have supervised nine PhD students (five currently active), ten MSc students (one currently active), and 18 undergraduates.

Shahar Mayo (2026 – 2028), TAU, Bioinformatics MSc (with Dr. Jérôme Tubiana, TAU)
Asaf Zorea (2024 – 2030), Life Sciences, TAU, PhD student
Ori Ben-Shir (2025 – 2030), Life Sciences, TAU, PhD student
Ella Rannon (2024 – 2029), Life Sciences, TAU, PhD student
Chen Agassy (2024 – 2026), TAU, Bioinformatics MSc
Nadav Shalev (2024 – 2026), TAU, Bioinformatics MSc
Guy Shur (2023 – 2028), Life Sciences, TAU, PhD student
Ori Ben-Shir (2023 – 2024), TAU, Bioinformatics MSc
Shirley Croitoru (2022 – 2027), Life Sciences, TAU, PhD student
Edan Gabay (2022 – 2025), TAU, Bioinformatics MSc
Maya Ben-Haim (2021 – 2024), TAU, Mathematical and Theoretical Biology MSc
Ella Rannon (2021 – 2024), TAU, Bioinformatics MSc
Adi Yanni (2020 – 2023), Life Sciences, TAU, PhD student (with Dr. Johann Elbaz, TAU)
Dr. Dan Mark Alon (2020 – 2023), Life Sciences, TAU, PhD student (with Dr. Gur Pines, AGRI)
Yatir Solan (2020 – 2023), TAU, Mathematical and Theoretical Biology MSc
Dr. Bruria Samuel (2019 – 2025), Direct PhD Track, Life Sciences, TAU, PhD student
Dr. Danielle Miller (2019 – 2024), Life Sciences, TAU, PhD student (with Prof. Adi Stern, TAU)
Dov Gertz (2019 – 2021), TAU, Bioinformatics MSc
Sagi Shaashua (2019 – 2021), TAU, Bioinformatics MSc

Additional personnel

Ortal Peretz (2022 – 2024), TAU, Bioinformatician
Yael Gershon (2022 – 2023), TAU, Lab Technician
Dr. Roy Rabinowitz (2022 – 2023), TAU, Researcher
Tomer Parket (2021 – 2024), TAU, Data Analyst
Ofir Arias (2020 – 2023), TAU, Software Specialist
Yuval Cohen (2020 – 2021), TAU, Programmer
Dr. Karin Mittelman (2018 – present), TAU, Research Associate

Teaching (since 2018)

- 2024 – 2025 Data Games, Co-Director, 2nd- and 3rd-year undergraduate and graduate student Project-Based Learning course, Life Science Faculty, TAU (~30 students)
- 2019 – 2025 Introduction to Bioinformatics, Co-Director, 2nd-year mandatory course, Life Science Faculty, TAU (~300 students)
- 2019 – 2025 Computational Genomics, Co-Director, 3rd-year and graduate student course, Computer Science School, TAU (~50 students)
- 2020 – 2023 Scientific Literacy, Director, 2nd-year mandatory course, Life Science Faculty, TAU (~20 students)

Teaching Excellence

Dean's Excellence Award and Top 100 Lecturers at Tel Aviv University, annually 2019–2024

Professional Service

Journals: Nature Microbiology, Nature Methods, Nature Reviews Microbiology, Nature Communications, Scientific Reports, Microbiome, Proceedings of the National Academy of Sciences (PNAS), FEMS Microbial Ecology, Bioinformatics, iScience, PLoS Computational Biology, Nucleic Acids Research (NAR), Molecular Biology and Evolution (MBE), The ISME Journal, Philosophical Transactions of the Royal Society B, mBio, PLoS One, GigaScience, The CRISPR journal, Genome Biology and Evolution (GBE), Protein Science

Funding Agencies: Dutch Research Council (NWO), Swiss National Science Foundation (SNSF), French National Research Agency (ANR), US-Israel Binational Science Foundation (BSF), Israel Science Foundation (ISF), Tel Aviv University Center for Combating Pandemics (TCCP), Koret-Berkeley-TAU (KBT)

- 2023 – present Board Member, The Israeli Society for Bioinformatics and Computational Biology (ISBCB).
- 2023 – present Edmond J. Safra Center for Bioinformatics, Steering Committee
- 2023 – 2025 Chair of the BSc. Bioinformatics program
- 2018 – 2023 Chair of the Life Science MSc. Bioinformatics program